

Dimensionality reduction

Material

https://youtu.be/QS6ZH_41b4s?si=fvaNGkqjslcxTEFn

- Making sense of [PCA](#)
- Understanding [t-SNE](#)
- [t-SNE explained](#) by Josh Starmer
- Understanding [UMAP](#)
- [Video](#) by one of the UMAP authors
- More info on [UMAP parameters](#)

Exercises

Load the seu dataset you have created yesterday:

Code starts here:

```
seu <- readRDS("day2/seu_day2-1.rds")
```

And load the following packages:

Code starts here:

```
library(Seurat)
```

Once the data is normalized, scaled and variable features have been identified, we can start to reduce the dimensionality of the data. For the PCA, by default, only the previously determined variable features are used as input, but can be defined using features argument if you wish to specify a vector of genes. The PCA will only be run on the variable features, that you can check with `VariableFeatures(seu)`.

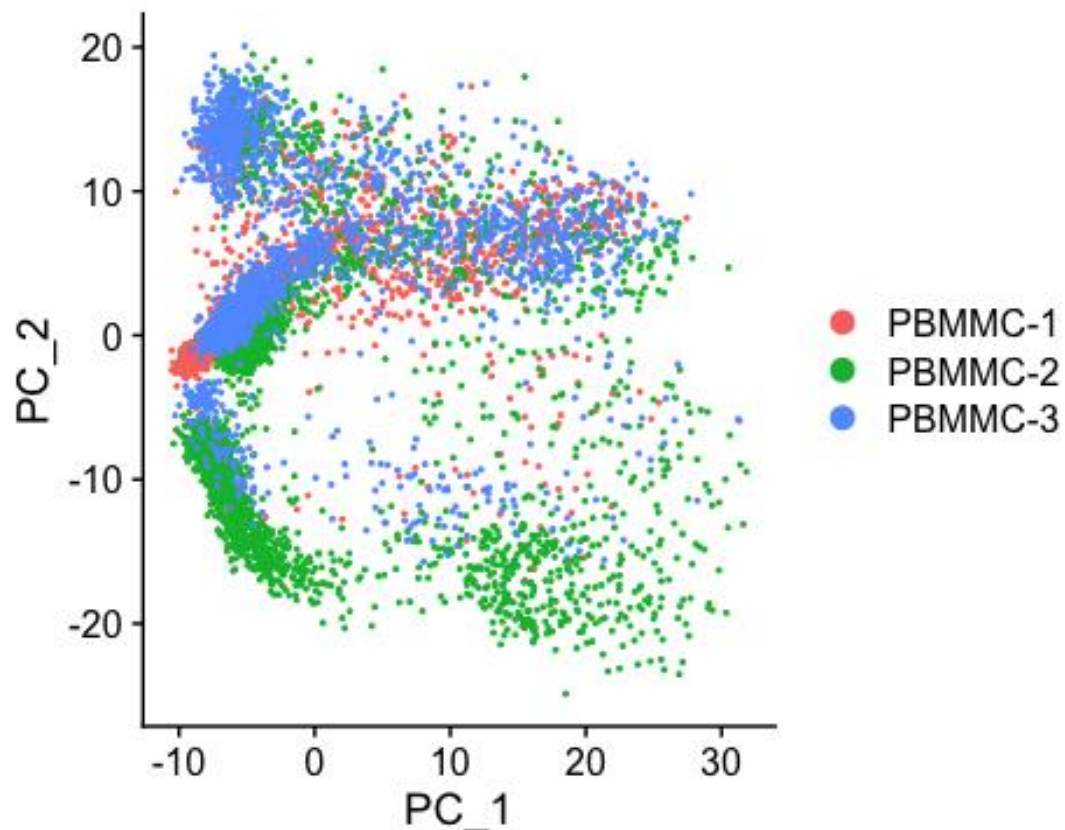
Code starts here:

```
seu <- Seurat::RunPCA(seu)
```

To view the PCA plot:

Code starts here:

```
Seurat::DimPlot(seu, reduction = "pca")
```



We can colour the PCA plot according to any factor that is present in `@meta.data`, or for any gene. For example we can take the column `percent.globin`:

Code starts here:

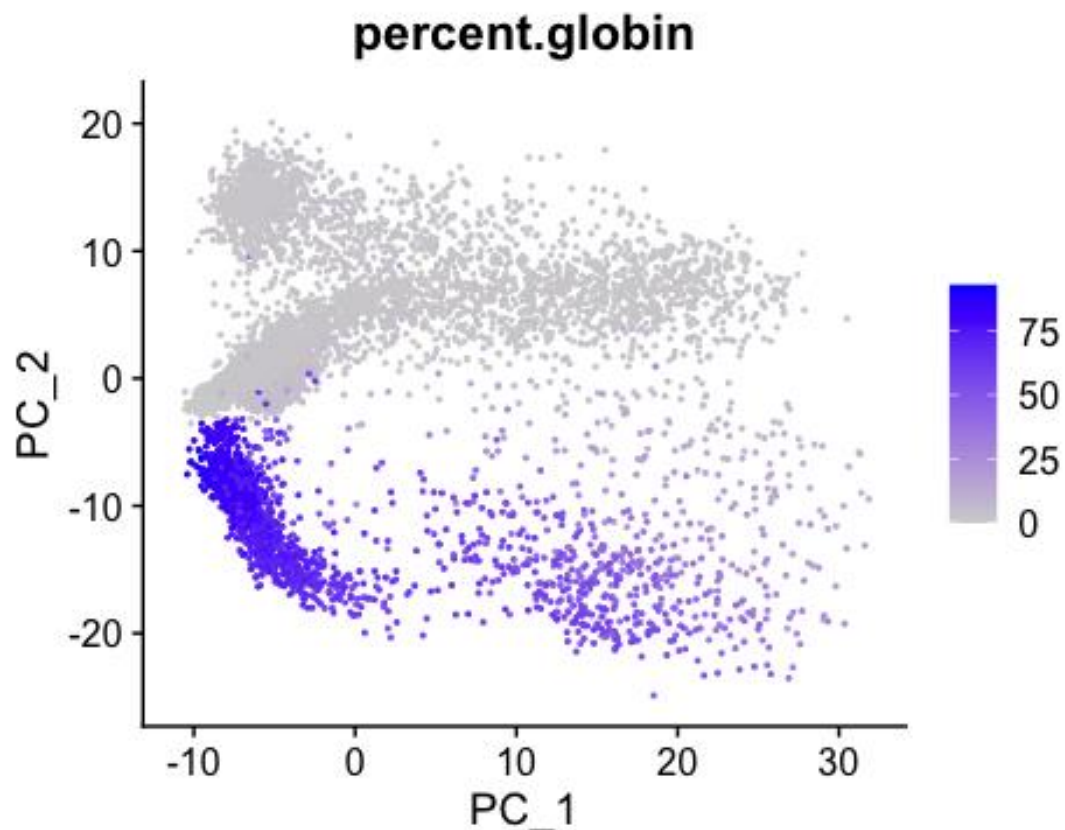
```
Seurat::FeaturePlot(seu, reduction = "pca", features = "percent.globin")
```

Warning: The ``slot`` argument of ``FetchData()`` is deprecated as of SeuratObject 5.0.0.

i Please use the ``layer`` argument instead.

i The deprecated feature was likely used in the Seurat package.

Please report the issue at <https://github.com/satijalab/seurat/issues>.



Note

Note that we used a different plotting function here: `FeaturePlot`. The difference between `DimPlot` and `FeaturePlot` is that the first allows you to color the points in the plot according to a grouping variable (e.g. sample) while the latter allows you to color the points according to a continuous variable (e.g. gene expression).

Exercise

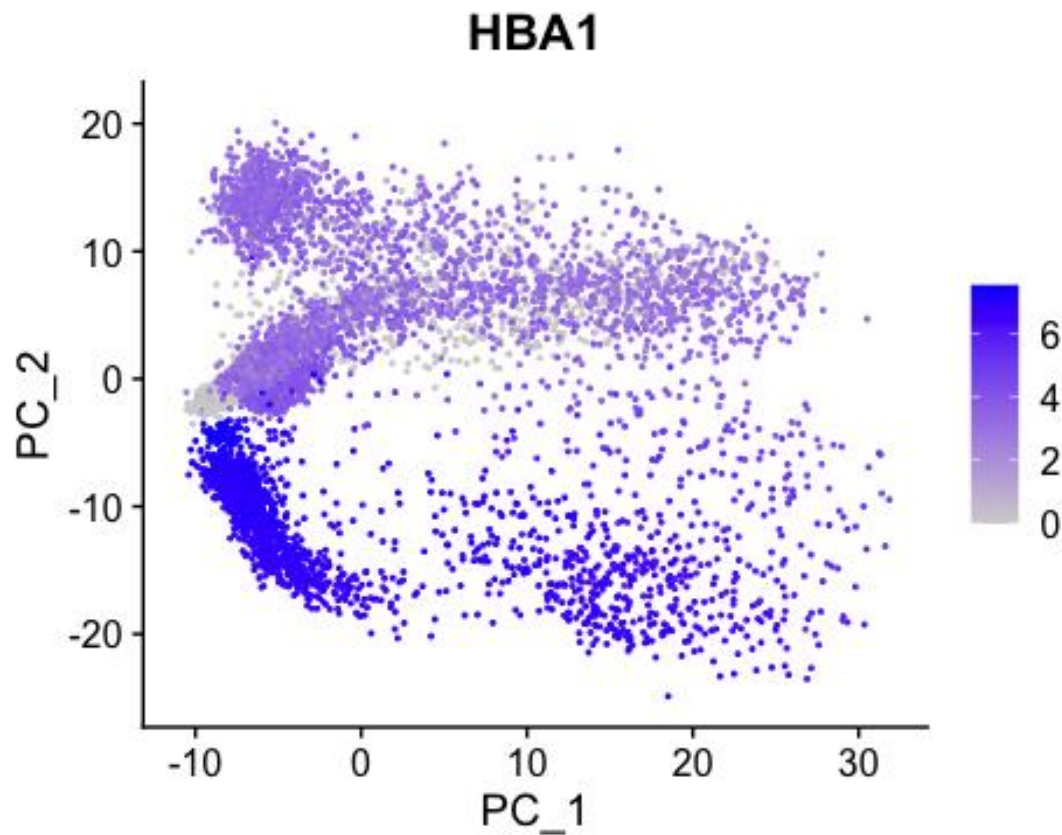
Generate a PCA plot where color is according to counts of a gene (i.e. gene expression). For example, you can take HBA1 (alpha subunit of hemoglobin), or one of the most variable genes (e.g. IGKC).

Answer

Generating a PCA plot coloured according to gene expression (here HBA1):

Code starts here:

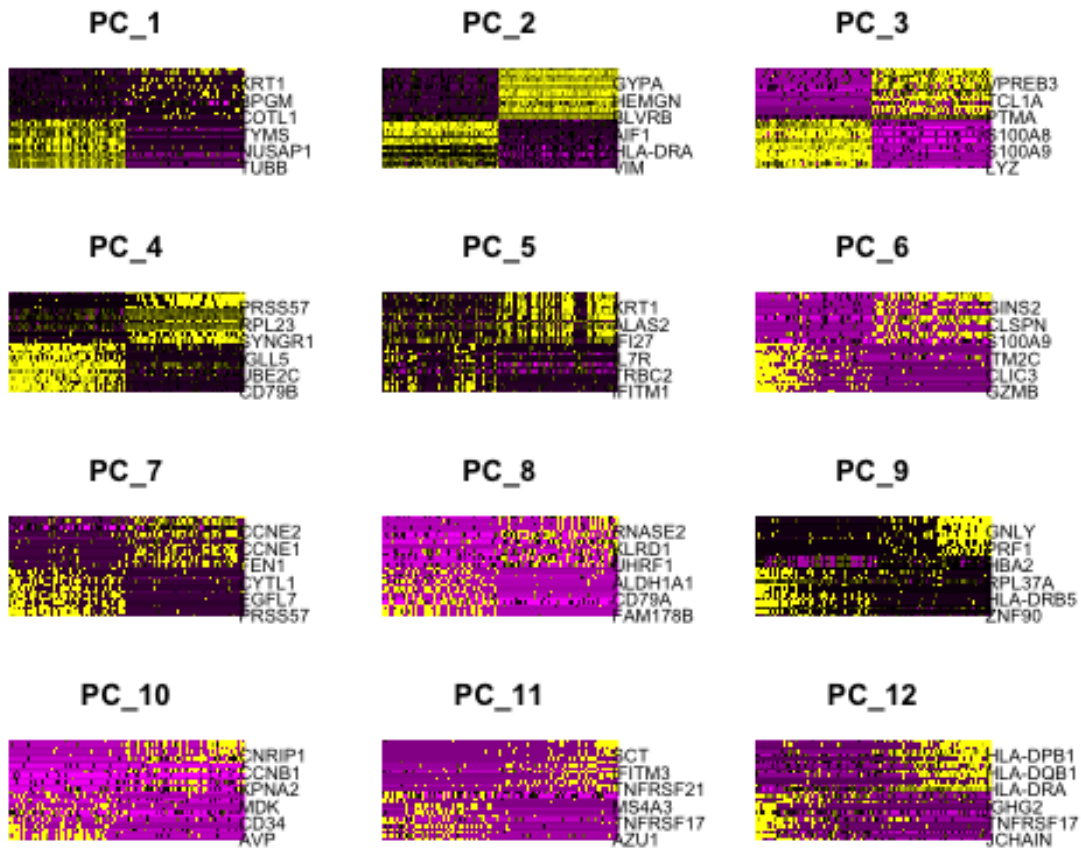
```
Seurat::FeaturePlot(seu, reduction = "pca", features = "HBA1")
```



We can generate heatmaps according to their principal component scores calculated in the rotation matrix:

Code starts here:

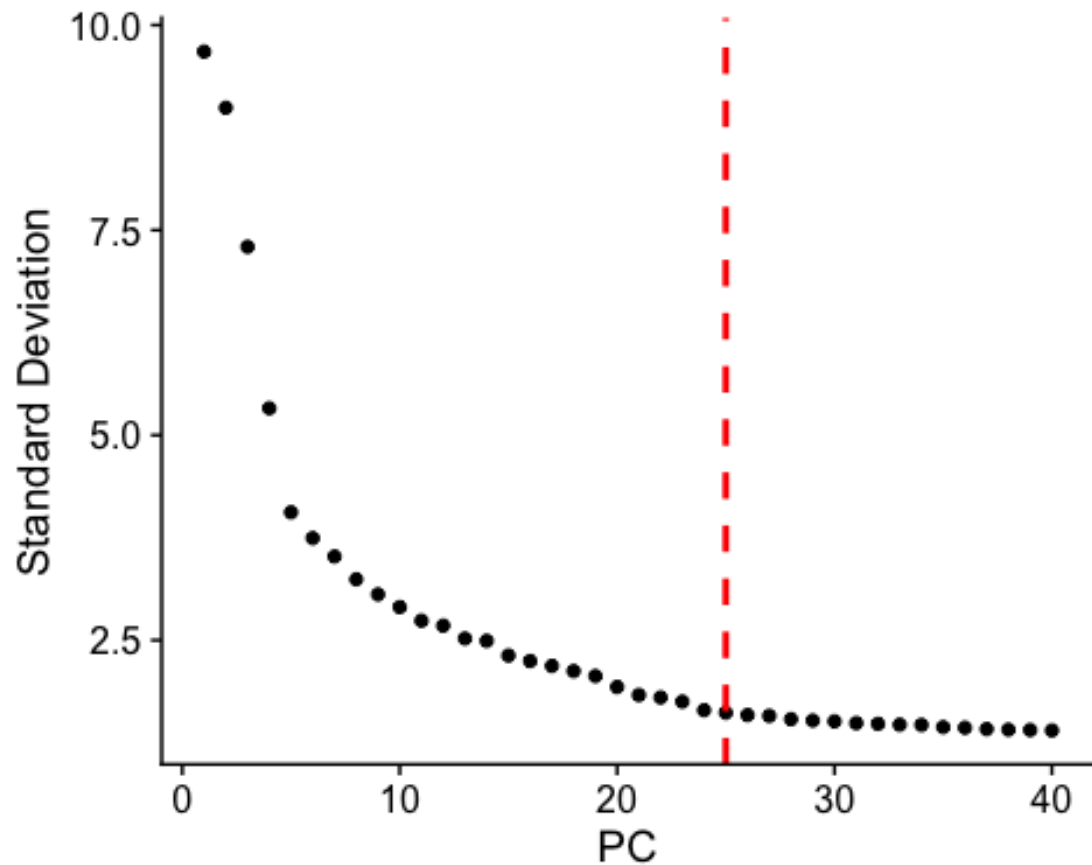
```
Seurat::DimHeatmap(seu, dims = 1:12, cells = 500, balanced = TRUE)
```



The elbowplot can help you in determining how many PCs to use for downstream analysis such as UMAP:

Code starts here:

```
Seurat::ElbowPlot(seu, ndims = 40) +
  ggplot2::geom_vline(xintercept = 25, linetype = "dashed", color = "red",
    linewidth = 1)
```



The elbow plot ranks principle components based on the percentage of variance explained by each one. Where we observe an “elbow” or flattening curve, the majority of true signal is captured by this number of PCs, eg around 25 PCs for the seu dataset.

Including too many PCs usually does not affect much the result, while including too few PCs can affect the results very much.

UMAP: The goal of these algorithms is to learn the underlying manifold of the data in order to place similar cells together in low-dimensional space.

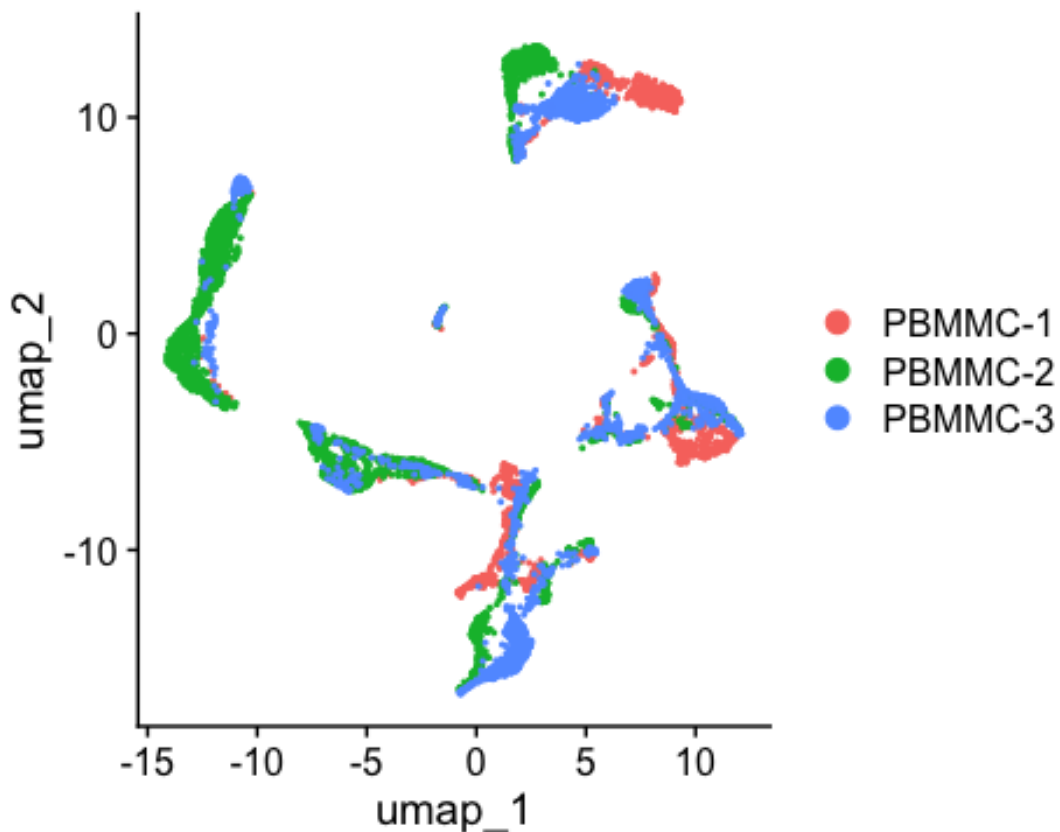
Code starts here:

```
seu <- Seurat::RunUMAP(seu, dims = 1:25)
```

To view the UMAP plot:

Code starts here:

```
Seurat::DimPlot(seu, reduction = "umap")
```



⚠ Exercise

Try to change:

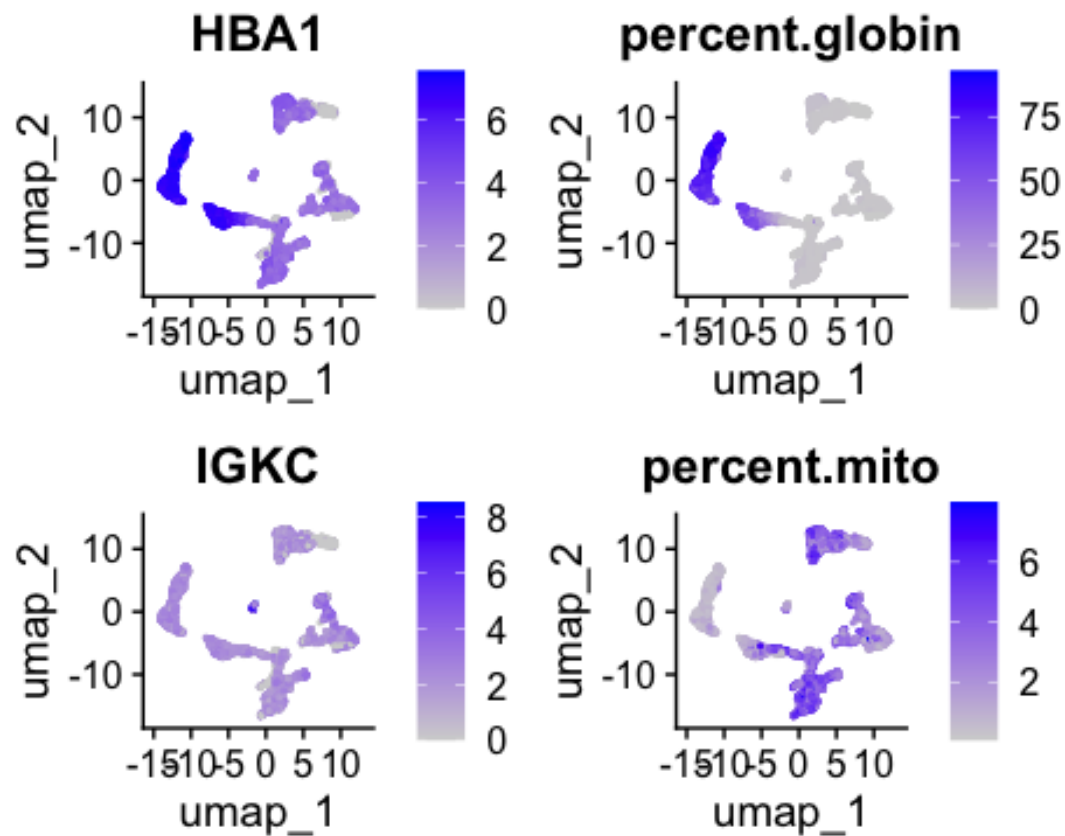
- A.** Color the dots in the UMAP according to a variable (e.g. `percent.globin` or `HBA1`). Any idea where the erythrocytes probably are in the UMAP?
- B.** The number of neighbors used for the calculation of the UMAP. Which is the parameter to change and how did it affect the output. What is the default ? In which situation would you lower/increase this ?
- C.** The number of dims to extremes `dims = 1:5` or `dims = 1:50` how did it affect the output ? In your opinion better few PCAs too much or too few ? Why does `dims = 1:100` not work? When would more precision be needed?

💡 Answer

Answer A

Code starts here:

```
Seurat::FeaturePlot(seu, features = c("HBA1",  
                                     "percent.globin",  
                                     "IGKC",  
                                     "percent.mito"))
```



The erythrocytes are probably in the cluster with a higher percentage of globin expression.

Answer B

Code starts here:

```
seu <- Seurat::RunUMAP(seu, dims = 1:25, n.neighbors = 5)  
Seurat::DimPlot(seu, reduction = "umap")
```



After having done these exercises, change the UMAP back to a UMAP based on the first 25 PCs, in order to replicate the exercises in the following chapters. Do this by:

Code starts here:

```
seu <- Seurat::RunUMAP(seu, dims = 1:25)
```